

Cybersecurity Concepts Lab for Teachers

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FIELD COPY

1



Lock It

PASSWORDS AND ACCOUNTS

2



Spot It

SCAMS AND FAKES

3



Share Smart

PRIVACY

4



Safe People

WHO TO TRUST

5



Kind and Well

KINDNESS AND BALANCE

6



Fix It

TELL, REPAIR, RECOVER

PLATE I. The six strands, hand-inked, in the order the lessons take them.

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OCTOBER 2026

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Demo 9: Data Footprints and Data Brokers (The Mosaic Puzzle)

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Status: DRAFT for teacher, counselor and legal review. Items marked [VERIFY] need checking against a primary source before publication.

AT A GLANCE

Big idea	Small, harmless facts add up. Put enough of them together and they point to one person: a "mosaic". Data brokers do this at scale. Your defense is to share less, set it tight, and think about where facts travel.
Grade bands	K-2 (footprints in the sand), 3-5 (Footprint Walk + mini puzzle), 6-8 (full Mosaic Puzzle + settings audit), 9-12 (full + brokers, re-identification research, law)
CSTA-style concepts	Impacts of Computing (privacy, tradeoffs between public and private), Data and Analysis, Networks and the Internet (cybersecurity) (codes: see Concepts Lab Index, Part 4 [VERIFIED 2026-10-05])
Time	15-minute and 45-minute versions
Cost and tech	Printed puzzle cards, stickers or paper footprints, tape, markers. No devices, no searching.
Teachers do it first	Yes. Part A about 20 minutes.

MAKE IT STICK

Try This Now (60 seconds, no prep): Meet "Quinn" (made up): plays soccer, goes to Maple Middle, has a dog named Biscuit. Pick two facts. Does the pair tell a stranger more than either fact alone? That is the mosaic effect. (Fictional person only; never look up a real one.)

01 The big idea

EVERY TIME YOU USE a phone, a game, a school tool, a store card or a smart speaker, you leave **data footprints**. Some you leave on purpose (a post, a photo, a username). Some are collected without you thinking about it (what you tapped, where your phone was, what you bought). Alone, one footprint rarely matters. **Together they form a picture**, which is the **mosaic effect**. Companies called **data brokers** gather, buy, match and sell such footprints to build profiles.

This lesson is not "the internet is terrible." It is: **you can understand how it works, make smart choices, and decide what to share.**

02 What a teacher should know (plain language)

2.1 ACTIVE AND PASSIVE FOOTPRINTS

KIND	EXAMPLES
Active (you share)	Posts, photos, comments, profile info, usernames, birthday shout-outs
Passive (collected as you go)	Location history, browsing and app activity, device identifiers, purchase history, search terms, loyalty-card scans
Shared about you (others post)	Tagged photos, team rosters, class lists, family posts
Inferred (guessed from the rest)	"Likely teen," "interested in sports," "household income band"

2.2 WHY THE MOSAIC MATTERS

- **Re-identification.** Removing names from data does not make it anonymous. A well-known study by Latanya Sweeney found that a large majority of the US population could be uniquely identified using just ZIP code, birth date and gender (Sweeney's paper reports 87% of the 248 million people counted in the 1990 census were likely unique on 5-digit ZIP code, gender and date of birth; Golle's 2006 re-analysis found about 61% for 1990 and 63% for 2000 data) [VERIFIED 2026-10-05: dataprivacylab.org/paper; crypto.stanford.edu/~pgolle].
- **Famous cases.** AOL's 2006 release of "anonymous" search logs let reporters identify a person from their searches; a 2008 study (Narayanan and Shmatikov, IEEE S&P) showed that "anonymous" movie ratings could be matched to public profiles [the 2008 paper is VERIFIED 2026-10-05; the AOL 2006 case is widely reported but was not checked].
- **Location is especially revealing.** A handful of location points can show where a person lives, goes to school and spends their weekends [the 2013 "Unique in the Crowd" study in Scientific Reports supports this:

VERIFIED 2026-10-05: Crossref record; its exact percentages were not checked, so none are claimed].

THE POINT FOR TEACHERS: privacy is not just secrecy; it is also about combining. "It's already public" does not mean "it's harmless to combine."

2.3 WHAT DATA BROKERS ARE

Data brokers are companies whose business is collecting, packaging and selling or licensing information about people. They draw from:

- Public records (property, court, voter files where permitted).
- Commercial sources (purchase data, loyalty programs, subscriptions).
- Apps and websites that share or sell data through advertising systems and software kits.
- Other brokers (data is resold).
- Inferences made by their own models.

They sell to marketers, lenders and insurers (subject to regulation), employers (screening), political campaigns, researchers, and **"people-search" websites** where anyone can look up a person's address and relatives. Some of this is regulated; much of the US market is lightly regulated [VERIFY current federal and state rules].

2.4 RULES AND RIGHTS (US-ORIENTED; CHECK YOUR STATE)

- **COPPA** limits collecting personal information online from children **under 13** without parental consent [VERIFIED 2026-10-05: [ftc.gov](https://www.ftc.gov/coppa) COPPA rule page; amended rule effective June 23, 2025, compliance by April 22, 2026].
- **FERPA** protects student education records at schools that receive US Department of Education funds; parents' rights pass to the student at 18 [VERIFIED 2026-10-05: studentprivacy.ed.gov].
- **State privacy laws** (for example California's CCPA/CPRA, and a growing list of other states) give residents rights to know, delete and opt out of sale or sharing [VERIFY]. California's Delete Act and its single-request deletion tool for registered data brokers (the "DROP" platform): residents could use it from January 1, 2026, and brokers must begin processing requests on August 1, 2026 [VERIFIED 2026-10-05: coppa.ca.gov]. Some states require data broker registration [VERIFY].
- **EU GDPR** (for readers in Europe) and the UK equivalent [VERIFY].
- Honest limits: opting out one broker does not erase data held by others; new data keeps flowing. Deletion is a **reduce and maintain** activity, not a one-time fix.

2.5 DEFENSE TOOLKIT (WHAT STUDENTS AND FAMILIES CAN ACTUALLY DO)

1. **Share less.** Ask: "Does this app really need my location, contacts, microphone?" Set permissions to "only while using" or "off."
2. **Tighten settings.** Private accounts; limit who can tag you; turn off location in photos; review ad personalization.
3. **Use different identities for different roles** (separate emails for school, games, shopping) where appropriate and with parental guidance.
4. **Think before posting about others.** Ask permission; avoid posting schedules, school names, uniforms, street signs, license plates and house views.
5. **The Billboard Test.** "Would I be okay with this on a billboard on my street, read by anyone, forever?" **The Grandma Test.** "Would I be okay if my grandparent, my future boss and a stranger saw this?"
6. **Check old accounts** and delete ones you no longer use [VERIFY that deletion does remove data per provider policy]. For people-search opt-outs and privacy dashboards, parents can help; do not hand personal data to "removal" services you have not researched [VERIFY reputation and cost].
7. **Tell a trusted adult** if someone online seems to know too much about you, asks where you live or tries to meet.

2.6 COMMON MISCONCEPTIONS

MYTH	BETTER UNDERSTANDING
"I have nothing to hide."	Privacy is about control and safety, not guilt.
"It's just a username, no one can tell who I am."	Username can be linked to other facts.
"Anonymous data is safe."	Combining "anonymous" records can re-identify people.
"Deleting a post erases it."	Copies and screenshots may persist.
"Private account = private data."	Apps, platforms and friends can still see and share it.

03 Materials (zero to low cost)

- **Footprint Walk station cards** (6 stations) and paper "footprints" (cut paper feet or sticky notes) in 5 colors for 5 data types.
- The **Mosaic Puzzle cards**: cards 1-6 and 8 are on the student handout (all people, places and details are **fictional**); card 7 is below. Chart paper, markers, tape, a "Broker" table with tokens or sticky dots.

04 The Footprint Walk (movement game, grades 3-12)

Setup. Six stations around the room, labeled **School Day, Phone, Store, Game, Home Smart Speaker, Neighborhood**. Each station has a stack of paper footprints in colors for data type: yellow = who you are, blue = where you are, green = what you like, red = what you buy, purple = who you know.

Play. Each student gets a "day card" with six tasks, one at a station (for example, "Check the school app", "Buy a snack with a store card", "Play an online game"). At each station, the student takes the footprint(s) listed on the station card and tapes them to their shirt or a "trail sheet". After six stations, count the footprints. Compare trails.

Round 2: The Minimal Trail. Same tasks, but each station has a **privacy choice** (turn off location, use a guest checkout, pay cash, private mode, mute the mic). Choices that reduce data skip a footprint but may cost convenience (the station card says what is lost, for example "no recommendations"). Students try to finish all tasks with the **fewest footprints** and the **fewest annoying costs**. Discuss the tradeoffs (a good CSTA-style point: security and privacy versus convenience).

Round 3: The Broker. One student plays the **Broker** who buys footprints at the Broker table for tokens (the student chooses whether to "sell" their footprints, to reveal that most trails are sold automatically in real life: a "terms of service" card says "sold by default"). The Broker assembles "profiles" (collect all the footprints into a pile per student). Students compare what the Broker knows.

05 The Mosaic Puzzle (fictional person; grades 5-12)

Rule. The puzzle uses a **fictional character** called "Quinn". The goal is to learn what **kinds** of conclusions combinations allow, not to practice finding real people. Never use a real student's name, never "look someone up", and never search the internet during this activity.

Cards 1-6 and 8 (a soccer roster, a cousin's birthday post, a photo caption, a neighbor's post, a loyalty-program record, a public fitness-app map and a broker profile for the made-up Quinn) are printed on the student handout without commentary. Cut them apart and hand out 2 cards per group member, so no single student holds the whole picture. Groups must talk to assemble it.

Optional card 7 (grades 8-12 only; skip it if any student may find it close to home): *Source: a family group chat screenshot posted by a relative. "Quinn's mom works late shifts, so the house is empty until 8."*

Questions for groups (answer from the cards only; the handout carries versions of questions 3, 4 and 5, and all answers are on the teacher page):

1. What do we know about Quinn's age and school? 2. Where might Quinn spend time? 3. What routine can we guess? 4. Which card gives the most **risky** detail, and why? 5. Which single card, if removed, breaks the most connections? 6. What would Quinn or Quinn's family change to reduce risk?

Facilitator framing: the lesson is not "never post." It is "be aware of what combines, and give extra care to location, schedules and who is home." Point out the good parts too: Quinn has a friendly community. Use the **defense lens** throughout: we are asking what a family should change, never how to find someone.

Follow-up: the Fix-It Round. Groups write "safer versions" of the cards. For example, "Happy birthday, Quinn!" (without the date or school), "After practice!" (no place), a fitness map set to private.

06 Part A: Teachers do it first (about 20 minutes)

- 1. Personal footprint inventory (8 min).** On paper only, list five things you share on purpose, five collected automatically and five others post about you. Do not name accounts or share the list.
- 2. Mosaic Puzzle (7 min).** Do the Quinn puzzle in groups of three. Which card did you *not* think of as sensitive?
- 3. Settings check (5 min).** On your own phone, reduce *one* permission (for example, location for one app). Do not compare with colleagues.

07 Part B: Teaching it

15-MINUTE VERSION

MINUTES	WHAT HAPPENS
0-2	Hook: "Footprints in the sand." Show that walking leaves trails.
2-8	Mini Footprint Walk (three stations: School, Phone, Store).
8-13	Mini Mosaic: four cards (1, 3, 4, 6) in groups; what did we learn?
13-15	Billboard Test exit slip.

45-MINUTE VERSION

MINUTES	WHAT HAPPENS
0-5	Hook and vocabulary (active, passive, inferred).
5-17	Footprint Walk Rounds 1 and 2.
17-30	Mosaic Puzzle with Fix-It Round.
30-37	Broker round and discussion of how brokers combine data; rights and opt-outs overview.
37-43	Settings audit (students write a "my family's top 3 changes" plan, to discuss at home).
43-45	Exit ticket and handout.

08 Teacher script

Hook (2 minutes). "Walk across this sandbox. Did you leave footprints? Online, we do too. One footprint tells little. But what if I collected a hundred? Today we'll find out how a picture of a person appears from tiny pieces, and what we can do about it. We use a made-up person named Quinn. We do not look up real people, ever."

During the Mosaic Puzzle. "No card here is 'secret' by itself. Which two together say something new? That's the mosaic."

On brokers. "Data brokers are companies whose business is combining footprints. Most of what they have come from sources people didn't think of as 'sharing'. Some places let you ask them to delete your data. It's a bit like weeding a garden: you have to keep at it."

Close. "You can't live without leaving footprints, and you don't have to. You can choose what you leave. Think: Billboard Test. Ask permission before posting about others. If anyone ever seems to know too much about you, or asks where you live, pause, and tell a trusted adult. It's never your fault."

09 Kid-friendly analogies by band

BAND	ANALOGY AND ACTIVITY
K-2	Footprints in the sand. Walk through a sand tray or on chalk; look at your trail. "Some things are okay to share with people we know; some things stay home: our full name, where we live, our school." Play "Share or Keep?" with picture cards.
3-5	The trail of crumbs. Hansel and Gretel crumbs (not a scary story; use a "bread trail" picture). Footprint Walk with 3 stations; what do crumbs tell you?
6-8	The jigsaw puzzle. Each piece alone is boring; together it is a picture. Full Mosaic Puzzle with Fix-It Round; settings audit.
9-12	Re-identification and the economics of data. Sweeney's study, linkage attacks as a concept, data broker business models, privacy law, consent and "notice and choice" limits. Evaluate: would a better law or a better design help?

10 Discussion questions

1. Which card in the puzzle seemed harmless but mattered most when combined? Why is location data so sensitive?
2. Is it fair that a company can sell facts about you? Who benefits? Who might be harmed?
3. What tradeoffs did you make in the Minimal Trail? Were they worth it?
4. (9-12) Should "anonymized" data be treated the same as personal data? Why or why not?

11 Level-up challenge

- **Design a privacy-friendly app** for a school club that works with the least data possible, explaining every field you ask for.
- **Footprint audit with a family member.** At home, with a parent, check **one** app's permissions and one account's privacy settings. Report only that you did it, not what you saw.
- **Law comparison (9-12).** Compare two privacy laws (for example, COPPA and a state law) on who is protected, what rights exist and what is not covered [VERIFY].

12 Safety and ethics notes (defense only)

- **Never apply the mosaic to a real person.** No searching for classmates, teachers or strangers. No "can you find my address?" challenges. This is the most important rule in this demo. If a student asks to try on themselves, say: "Let's do it at home with a grown-up and only look at your own settings."
- **No OSINT how-to.** Do not teach tools or methods for finding people. Teach what *makes people findable* so students can reduce it.
- **Do not ask students to share their own private information** (address, schedule, family details). Use the fictional puzzle.
- **Sensitive circumstances.** Some students may have safety reasons to be hard to find (for example, custody or domestic-safety situations). Keep the discussion general; offer private follow-up; follow school policy.
- **Disclosures.** If a student says a stranger knows where they live or is contacting them, or that someone is tracking them, follow your reporting process (see the Index box). Do not ask for details in front of the class.

13 **Assessment idea**

- **Exit ticket.** "Name two harmless facts that become risky together. Name one change to reduce the risk." Rubric: 3 = identifies mosaic and one action; 2 = identifies risky facts but no mosaic; 1 = "don't post anything".
- **Performance task.** Give a fictional social media profile (printed). Students highlight data that could combine to reveal location or routine and propose edits.

14 **Sources and [VERIFY] list**

- Latanya Sweeney, "Simple Demographics Often Identify People Uniquely" (2000); Golle, "Revisiting the Uniqueness of Simple Demographics in the US Population" (2006) [VERIFIED 2026-10-05: both papers read for the figures above].
- Narayanan and Shmatikov, "Robust De-anonymization of Large Sparse Datasets," IEEE Symposium on Security and Privacy, May 2008 [VERIFIED 2026-10-05: Crossref record].
- de Montjoye et al., "Unique in the Crowd: The privacy bounds of human mobility," Scientific Reports, March 2013 [VERIFIED 2026-10-05: Crossref record].
- COPPA (FTC), FERPA (US Department of Education) [VERIFIED 2026-10-05: ftc.gov; studentprivacy.ed.gov].
- California Delete Act and DROP platform [VERIFIED 2026-10-05: cppa.ca.gov]; other state data broker registries [not checked].
- FTC report "Data Brokers: A Call For Transparency and Accountability" (May 2014) [VERIFIED 2026-10-05: ftc.gov]; the 2024 FTC location-data actions were not verified.
- UKCIS Education for a Connected World, Privacy and Security strand [VERIFY].

Student Handout: Footprints and Mosaics

Name: _____ (do **not** write your real address or schedule here)

Footprint types

TYPE	EXAMPLE
Active	A post I share
Passive	Where my phone is
Shared about me	A tagged photo
Inferred	"Likes soccer" guessed from my buys

The Mosaic Puzzle (Quinn is a made-up person)

CARD	SOURCE (MADE UP)	FACT
1	School soccer roster (public)	"Quinn A., Grade 7, Maple Middle School, number 9"
2	Birthday post by a cousin	"Happy 13th, Quinn! Can't wait for the cake on March 14!"
3	A photo caption	"After practice at the park by the blue bridge. Tuesdays and Thursdays are the best!"
4	Neighbor's post	"Biscuit the dog says hi from the red door on Elm Street."
5	Store loyalty program	"Frequent purchases: size 6 cleats, peanut-free snacks, manga volumes 1-8."
6	Fitness app map (public by default)	"Runs 4:00 pm Tue/Thu from a spot near Elm Street to the park"
8	A data broker profile	"Household with a teen, middle income band, device: phone model X, interests: soccer, anime."

- Which card numbers **together** show Quinn's regular Tuesday routine?
____, _____, _____
- Which card is the riskiest, and why? ____

- If you took away one card, which would break the most connections?

Fix-It

Rewrite one card so it is safer:

Before: _____

After: _____

Billboard Test

BEFORE I POST: "WOULD I be okay with this on a billboard on my street, seen by anyone, forever?" Yes / No / Ask first

Three things I can do

1. Turn off location for _____ (ask a grown-up)
2. Set _____ to private
3. Ask a friend before I post _____

Remember

- Small facts add up. Location, schedules and who is home need extra care.
- Never look up real people for fun. Never post someone else's private info.
- If anyone online seems to know too much about you, **Pause. Check. Tell** a trusted adult. It is not your fault.

End of student handout.

TEACHER COPY ONLY:

Answers for Demo 9 (do not distribute to students)

FIELD NOTE

No. 1

WARNING: Teacher page. Remove before copying. The student handout ends on the page before this one. Print the handout pages only, or detach this page first. Do not post it, project it or send it home.

Group questions (from the cards only):

1. Age and school: about 13, Grade 7, Maple Middle (cards 1 and 2).
2. Where Quinn spends time: a park near a blue bridge; the Elm Street area (cards 3, 4, 6).
3. Routine: Tuesday and Thursday after practice around 4:00 (cards 3 and 6).
4. Riskiest: card 6 (routine and location); card 7 if used (the house is empty until 8).
5. Removing card 6 or card 3 breaks the most connections; removing both leaves only a neighborhood guess.
6. Reduce risk: make the fitness map private, do not post when the house is empty, do not tag the park, share birthdays without school or date.

Handout: 1 cards 3, 4 and 6 together (with 1 for who). 2 Card 6 (routine), or 7 if used. 3 Card 6 (or 3). **Fix-It examples:** "Happy birthday, Quinn!" (no date or school); "After practice!" (no place); a fitness map set to private.

What groups usually find: no single card is dangerous, but cards 3, 4 and 6 together point to a neighborhood and a routine, and optional card 7 shows when no adult is home. Skip card 7 if any student may find it close to home.